

Survival outcomes and prognosis predictors of uveal melanoma from Indian population

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Purpose: Choroidal melanoma is an uncommon malignancy in the Asian Indian population, with limited survival data available for this demographic. This study aims to evaluate the survival outcomes and prognostic factors in a cohort of Asian Indian patients with choroidal and ciliochoroidal melanoma. **Methods:** We conducted a retrospective analysis of 150 patients with choroidal and ciliochoroidal melanoma between 2011 and 2023 at a single tertiary eyecare and ocular oncology center in India. The demographic and clinical profile, using Collaborative Ocular Melanoma Study (COMS) and American Joint Committee on Cancer (AJCC) classification and histopathological features, were recorded. Patients were contacted through various avenues such as telephone, email, and other social media to know their survival status. Kaplan–Meier survival curves and Cox regression were used to determine survival probabilities and prognostic factors. **Results:** Of the 150 patients, nearly half (48%) of the patients were 50 years or younger at presentation. The melanoma metastasis-related mortality was 16%. The five-year survival was more than 90% in stages T1 and T2, 80% in stage T3, while it dropped to almost 65% in stage T4. The five year disease specific survival (DSS) was 84% (95% CI: 77%–90%), and the 10 year DSS was 66% (95% CI: 53%–77%). Multivariate Cox proportional hazards analysis revealed a significantly increased risk of disease-specific mortality with higher tumor stage, i.e., T2: hazard ratio (HR) = 1.78 (95% confidence interval [CI]: 1.01–3.13; $P = 0.045$), T3: HR = 4.24 (95% CI: 2.35–7.64; $P < 0.001$) and T4: HR = 6.38 (95% CI: 3.21–12.67; $P < 0.001$), compared to T1. Overall, 29 patients developed metastasis during the study period. The five-year metastasis rate was 16% (i.e., 83% of metastasis occurring within first five years), and 10-year metastasis rate was 19.3%. **Conclusion:** This study highlights favorable survival outcomes in Indian patients with choroidal melanoma, despite a younger age at presentation. The critical window for metastasis management is within the first five years, underscoring the necessity of structured follow-up protocols.

Key words: Asian Indian population, choroidal melanoma, mortality, prognosis, survival

Choroidal melanoma is the most common primary intraocular malignancy in adults but is rare in the Asian population. The reported incidence in Asians (0.25–0.64 per million) is significantly lower than in the United States (5 per million) and Europe (2–8 per million). Additionally, Indian patients often present at a younger age compared to their Western counterparts.^[1–5]

The prognosis of choroidal melanoma is based on clinical, anatomic, histopathologic, and genetic profile factors. Older age, male gender, larger tumor basal diameter and thickness, ciliary body location, diffuse tumor configuration, extraocular tumor extension, epithelioid cell type, high mitotic activity, higher microvascular density, extravascular matrix patterns, tumor-infiltrating lymphocytes and macrophages, chromosome 3 loss, 8q gain, 1p loss, and 6q loss are the various poor prognostic factors.^[6] Various Western studies, notably the Collaborative Ocular Melanoma Study (COMS), have provided benchmarks for survival outcomes.^[7–10] However, data specific to the Asian population is sparse.^[11–14] So far, very few Indian

studies have been published on the clinicopathological profile of choroidal melanoma and the striking differences with Caucasian cohorts, but survival outcomes, particularly in the Indian population, have not been clearly studied yet.^[4,5,15,16]

Also, genetic markers have now proven to predict prognosis. The cancer genome atlas (TCGA) based on the DNA alterations categorized uveal melanoma into four prognostic groups: Group A (Disomy 3 with extra 6p) with favorable outcome, Group B (Disomy 3 with extra 6p and partial 8q gain) with late metastasis, Group C (Monosomy 3 with extra 8q), and Group D (Monosomy 3 with multiple copies of extra 8q) with unfavorable outcomes.^[17] More recently, the Collaborative Ocular Oncology Group (COOG) also proposed that the 15 gene expression profiling (GEP) combined with PRAME (PReferentially expressed Antigen in Melanoma) expression status has a superior prognostic accuracy of metastasis-free survival over clinical prognostic variables or GEP alone.^[18]

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This study aims to evaluate the survival outcomes and prognostic factors in a cohort of Asian Indian patients with choroidal and ciliochoroidal melanoma comparing with Western population.

Methods

Study design and setting

This retrospective study included patients diagnosed with choroidal or ciliochoroidal melanoma at the Ocular Oncology Department of a tertiary eye care center between 2011 and 2023. This study protocol was reviewed by the Institutional Ethics Board, and the need for approval was waived off due to the retrospective nature of the study.

The demographic and clinical profile, including geographic origin of the patient, age, gender, contact details, symptoms at onset of presentation, change in visual acuity, clinical features, diagnostic workup, treatment modalities, and duration of follow-up, was collected retrospectively from the digital health records. Clinical features evaluated were tumor location, involvement of optic nerve/ciliary body, orange pigment, drusen, surface hemorrhages on the tumor, subretinal fluid/exudative retinal detachment (RD), and vitreous hemorrhage (VH). Tumor was classified into small, medium, and large size based on COMS Group criteria and also based on AJCC TNM classification system (8th edn).^[19,20] Diagnostic workup included ultrasound imaging for tumor configuration, height, largest basal diameter, magnetic resonance imaging (MRI), liver function tests, abdominal ultrasound, chest radiography, and fine needle aspiration biopsy reports (if available). Treatment protocol was brachytherapy (Ru-106 plaque with target apex dose – 100 Gy)

with or without adjuvant transpupillary thermotherapy for small and medium-sized tumors and primary enucleation for large-sized tumors. If the tumor size progressed despite brachytherapy after periodic follow-ups, secondary enucleation was performed. Patients who developed radiation retinopathy and macular edema were treated with anti-VEGF (vascular endothelial growth factor) injections. Eyes with toxic tumor syndrome were treated with vitrectomy and tumor endoresections. Treatment details were recorded, including the type of treatment modality, number of treatment sessions (single/multiple), and the histopathologic features of enucleated eyes. Patients were followed up regularly until the tumor regressed and then periodically once in six-months, once stable. Metastatic surveillance was done by liver function tests every six-monthly and chest X-ray and abdominal ultrasound every yearly.

The patients were contacted through telephonic conversations, emails, WhatsApp, and other social media platforms to collect details about patients' survival status, history of confirmed metastatic disease, current and systemic treatment condition, and duration and cause of death in case of death. The cause and time of death were noted. Patients who were not traceable were excluded from the study. Patients who either passed away due to metastasis within the first year of diagnosis or completed a minimum follow-up of one year duration were included in the study. Clinicopathological features were correlated with mortality and metastasis rates to assess the outcome.

Statistical analysis

Data was recorded and coded in Microsoft Excel™ and analyzed using JASP™ (Version 0.14). Descriptive statistics (mean, median, and standard deviations [SD]) were used for continuous variables,

Table 1: Tumor characteristics and classification based on AJCC staging and COMS staging

AJCC T stage	Count	AJCC 8 classification	COMS	Tumor characteristics
T1a	21	T1-24 (16%)	Small	Drusen (2 eyes)
T1b	2		6 (4%)	Orange pigment/Lipofuscin (9 eyes)
T1c	1			SRF/Exudative detachment (9 eyes)
T2a	43	T2-46 (31%)	Medium	Drusen (11 eyes)
T2b	3		79 (53%)	Orange pigment/Lipofuscin (8 eyes)
				SRF/Exudative detachment (24 eyes)
T3a	39	T3-60 (40%)	Large	Drusen (2 eyes)
T3b	13		65 (43%)	Orange pigment/Lipofuscin (4 eyes)
T3c	4			SRF/Exudative detachment (45 eyes)
T3d	4			
T4a	9	T4-20 (13%)		Drusen (1 eye)
T4b	7			Orange pigment/Lipofuscin (2 eyes)
T4c	1			SRF/Exudative detachment (15 eyes)
T4d	1			
T4e	2			

Table 2: Management modalities of tumor based on COMS staging

	Single session of Brachy therapy	Multiple treatments (sandwich therapy, multiple transpupillary thermotherapy, repeat brachytherapy)	Secondary enucleation	Primary enucleation	Deferred treatment
Small (6 eyes)	4	2	1	-	
Medium (79 eyes)	63	24	11	8	
Large (65 eyes)	2	-	1	61	2

AJCC - American Joint Committee on Cancer, COMS - Collaborative Ocular Melanoma Study

and frequencies and percentages for categorical variables. Group comparisons for continuously distributed data were made using an independent sample 't' test, and non-parametric tests in the form of the Chi-square test and Mann-Whitney U test for categorical variables were performed. Similarly, linear correlation between two continuous variables was analyzed using Pearson's correlation test for normally distributed data, and Spearman's correlation test was used for non-normally distributed data. Age categorization was done in decade groups (e.g., 21–30 years and 31–40 years) to assess metastasis risk across age groups, with significance tested using the Chi-square test. For predicting factors affecting the survival outcome, the Log Rank test was used for categorical variables and Cox proportional hazards (Cox-PH) regression for continuous variables. Hazard ratio (HR) and 95% confidence intervals (CIs) were calculated. Cumulative survival probability plots have also been analyzed using Kaplan–Meier curves, including overall survival, five-year survival, and survival in patients with metastasis.

Results

Demographics

Among 167 treated choroidal melanoma patients (2011–2023), 150 were included, with 87 males (58%). Seventeen patients were excluded due to untraceable survival status. The mean and median follow-up durations were 66 months and 61 months, respectively (range: 5–191 months), with a mean age at presentation of 51.2 years (range: 20–96 years). A total of 47% of patients were under 50, and 51% had right-eye involvement. Notably, one patient developed melanoma during pregnancy, while four patients had Nevus of Ota, and four had iris involvement, including one with heterochromia iridis.

Tumor characteristics

Of the 150 eyes, nearly half of the patients (52%; 78 eyes) had medium-sized tumors (COMS classification), and around 40%

had T3 tumors (AJCC classification). Table 1 enlists the tumor staging. The mean basal diameter was 12.46 ± 4.22 mm, and the mean tumor height was 8.02 ± 3.73 mm. Younger age was linked to a larger tumor height ($P = 0.038$).

Posterior pole involvement was the most common tumor location, seen in 65 eyes (43%), with 36 eyes (24%) involving the fovea and 22 eyes (15%) showing optic nerve head (ONH) obscuration. Diffuse involvement was noted in 11 eyes (7%). Tumor extended to the ciliary body in 31 eyes (21%) and iris in four eyes (2.6%). VH occurred in 14 eyes (9%), with vitreous seeding in two eyes (1.3%) and anterior chamber seeding in one eye (0.7%). Histopathologic data were analyzed for enucleated eyes and are discussed in the subsequent management section.

Management

Among the 150 eyes analyzed, 69 eyes (46%) underwent primary enucleation, 13 eyes (8.7%) underwent secondary enucleation, 71 eyes (47%) received Ru-106 plaque brachytherapy (of which two eyes received repeat brachytherapy), 24 eyes (16%) were treated with transpupillary thermotherapy (TTT) as primary (two eyes – 1.3%) or adjunct or sandwich therapy, and two patients refused enucleation (due to metastatic disease at presentation). Nearly 80% of patients underwent post-treatment metastatic surveillance.

Three eyes underwent diagnostic biopsy for atypical tumors, and three eyes underwent endoresection post-brachytherapy for toxic tumor syndrome (at a mean follow-up – 22.5 months).

Globe salvage rate was 45%, with 46% (69 eyes) primary enucleations, 8.7% (13 eyes) secondary enucleations, and [Table 2]. Histologically, spindle cell tumors were most common (47%),

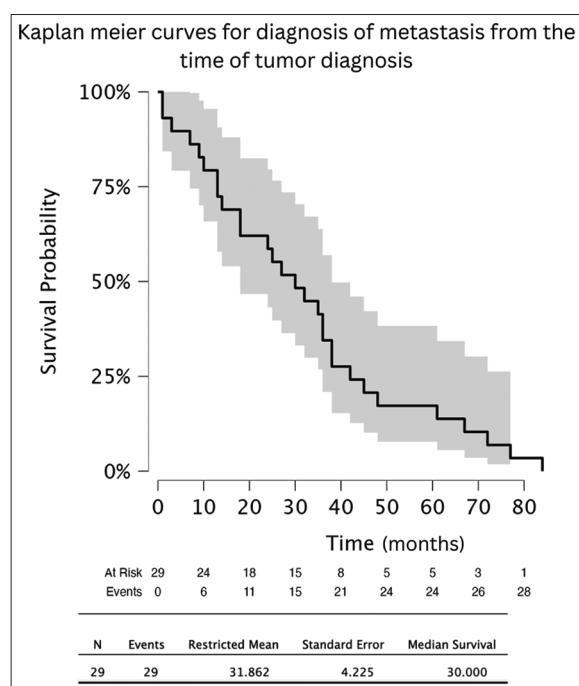


Figure 1: KM curves for diagnosis of metastasis from the time of tumor diagnosis

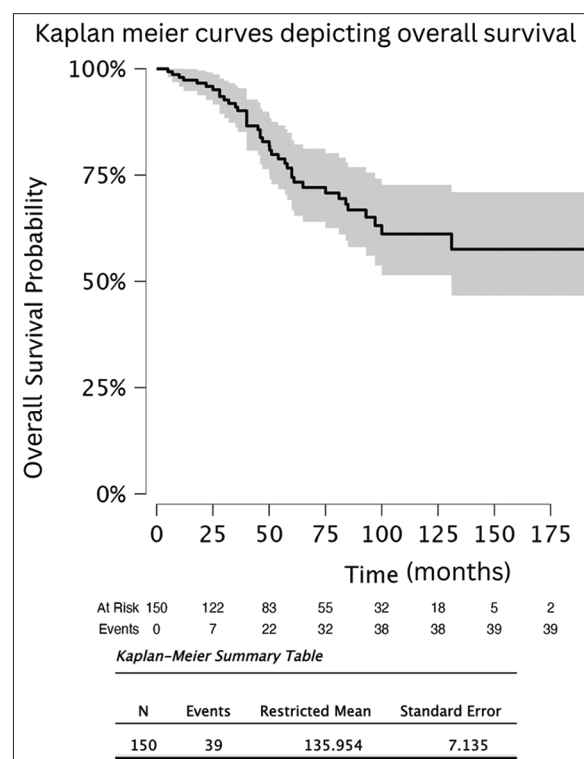


Figure 2: Overall survival probability curves

Table 3: Characteristics of patients with metastasis related deaths

Patient	Age (Years)	Gender	Fovea Involved	ONH Involved	Ciliary Body Involved	Sclera involved	Anterior chamber involved	Metastasis At Presentation	AJCC T STAGE	COMS Tumor classification	TNM HPE GRADING	Tumor Recurrence	Location Of Metastasis	Survival after diagnosis in years	Diagnosis of mets after tumor diagnosis in years
1	33	Male	No	Yes	No	No	No	No	T3a	Large	Mixed		Liver	10.9	7.1
2	41	Female	No	No	No	Yes	No	No	T4c	Large	Spindle		Liver, Lung	7.1	6
3	29	Male	No	No	No	No	No	No	T2a	Medium	Spindle	Yes	Liver	6.3	5.6
4	65	Female	No	Yes	No	No	No	Yes	T3a	Large			Bone	0.4	0.1
5	49	Male	Yes	No	No	No	No	No	T2a	Medium			Liver	3.9	3.2
6	21	Male	No	No	Yes	Yes	No	No	T3d	Large	Spindle		Liver	3.3	3
7	53	Male	No	No	Yes	No	No	No	T3b	Medium	Spindle		Liver, skin	3.3	1.5
8	63	Male	Yes	No	No	Yes	No	No	T1c	Medium	Mixed	Yes	Bone	5.4	5.1
9	55	Female	No	No	No	Yes	No	No	T3c	Large	Spindle		Diffuse	5.0	3
10	56	Male	Yes	No	No	No	No	No	T4a	Large	Epithelioid		Liver, Lung	1.5	0.9
11	64	Male	No	Yes	Yes	Yes	No	No	T3d	Large	Mixed		Liver	3.0	1.1
12	38	Male	No	Yes	No	No	No	No	T3a	Large	Spindle		Liver	2.4	1.5
13	45	Female	No	No	Yes	Yes	No	No	T4e	Large	Mixed		Diffuse	2.9	0.8
14	20	Female	Yes	No	No	No	No	No	T3a	Medium			Liver	4.9	3.2
15	43	Female	Yes	No	No	No	No	No	T2a	Medium			Lung	8.1	6.4
16	45	Female	Yes	Yes	Yes	Yes	No	No	T4e	Large	Epithelioid	Yes	Liver, Bone	4.2	2.1
17	45	Male	No	No	Yes	No	No	No	T3b	Large	Epithelioid		Liver	4.3	2.3
18	80	Male	Yes	No	No	No	No	No	T3a	Large	Spindle		Liver	2.4	1.2
19	57	Female	No	No	No	Yes	No	No	T3c	Large	Spindle		Liver	4.2	3.8
20	46	Male	No	No	No	No	No	No	T3a	Large			Liver	4.5	4
21	70	Female	No	No	No	No	No	No	T2a	Medium			Liver	4.8	3.5
22	50	Female	No	No	No	No	No	No	T3a	Large	Spindle		Liver	3.8	2
23	77	Male	No	No	No	Yes	No	No	T3c	Large	Mixed		Liver	3.0	2.7
24	50	Male	No	No	No	No	No	No	T4a	Large	Mixed		Liver	4.0	2.5
25	71	Female	No	No	No	No	No	No	T4a	Large	Mixed		Liver	3.8	2.9
26	48	Female	No	No	No	No	Yes	No	T3a	Large	Mixed		Bone	0.6	0.3
27	73	Female	Yes	No	Yes	Yes	Yes	Yes	T3d	Large	Epithelioid		Brain	3.3	0.1
28	29	Female	Yes	Yes	Yes	Yes	Yes	No	T4d	Large	Mixed		Diffuse	0.8	0.6
29	38	Male	No	Yes	No	Yes	No	No	T2c	Medium	Epithelioid	Yes	Liver, Lung	1.8	1.1

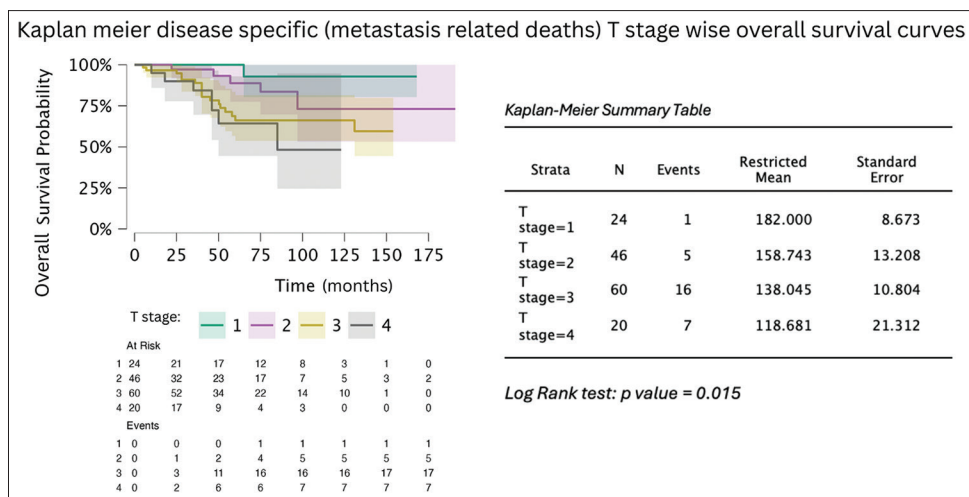


Figure 3: Kaplan–Meier disease-specific (metastasis-related deaths) T-stage-wise overall survival curves

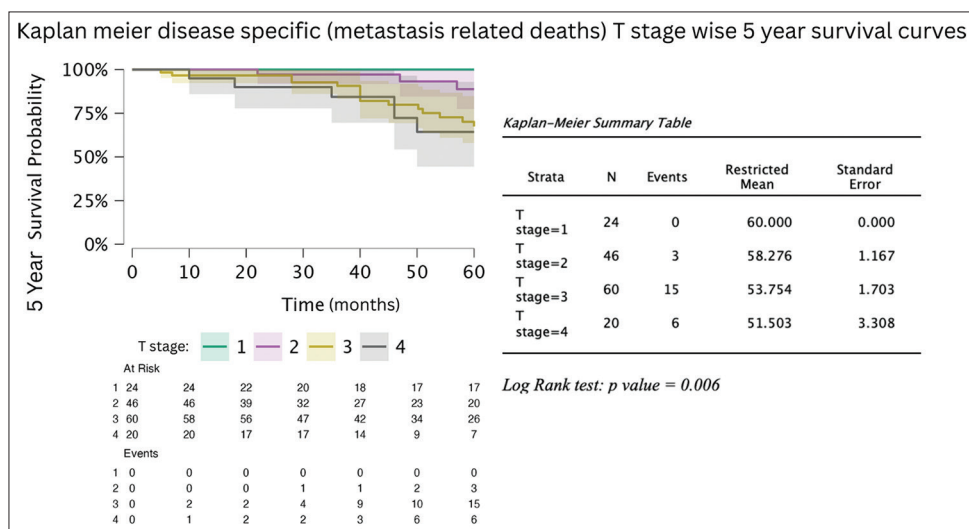


Figure 4: Kaplan–Meier disease-specific (metastasis-related deaths) T-stage-wise five-year survival curves

followed by mixed (39%) and epithelioid (14%). Among spindle cells, type B was predominant (73.5%). Scleral involvement was noted in 30% of enucleated eyes, with half showing partial (11 eyes) and half full (10 eyes) thickness involvement. Tumor recurrence was observed in 13 eyes (8.7%) post-brachytherapy and TTT treatment.

Metastasis

Of the 150 patients, 29 (19.3%) developed distant metastasis during follow-up. Twenty-four of these 29 patients (83%) developed metastasis within the first five years of their initial diagnosis, consistent with known high-risk periods. Six patients (4%) developed metastasis within the first year, and two patients (1.3%) had detectable metastasis at initial presentation. The mean time from tumor diagnosis to confirmed metastasis was 2.56 years (Range: 0.1–7 years). A Kaplan–Meier (KM) curve illustrating survival probability is shown in Fig. 1. The overall and five-year metastasis-free survival KM curves in relation to T stage are presented in Figs. 2 and 3, respectively. The number of patients at risk is shown at relevant time points.

The liver was the most common site of metastasis (21 patients, 72%), followed by lung (10 patients, 34%), bone (five patients,

17%), skin (two patients, 7%), and brain (two patients, 7%). Multiple organ involvement was observed in six patients (21%). Table 3 demonstrates complete metastatic site distribution.

Metastasis management varied depending on the extent and timing. Few patients received systemic chemotherapy or immunotherapy, including dacarbazine, interferon, and checkpoint inhibitors, while others received only supportive or palliative care, or declined systemic treatment due to poor performance status or financial limitations.

Patients' age and risk of metastasis were analyzed by grouping them into decade-wise categories, i.e., 21–30 years, 31–40 years, 41–50 years, 51–60 years, 61–70 years, 71–80 years, 81–90 years, and 91–100 years. Age did not significantly influence metastasis risk ($P = 0.118$) [Table 4].

Outcome

In the eyes where the globe was salvaged, compared to the initial presenting visual acuity, vision remained stable in 34%, improved in 22%, and deteriorated in 44% of the eyes.

Post-treatment complications like radiation retinopathy were noted in seven eyes post-brachytherapy, which was treated

Table 4: Comparison of age-wise categories with risk of metastasis			
Age Categories	Metastasis		Total
	Absent	Present	
21-30 years	9	4	13
31-40 years	13	3	16
41-50 years	34	8	42
51-60 years	38	5	43
61-70 years	20	3	23
71-80 years	5	4	9
81-90 years	2	1	3
91-100 years	0	1	1
Total	121	29	150

Chi-square test: $P=0.118$

by anti-VEGF injections; three eyes developed toxic tumor syndrome, for which vitrectomy with tumor endoresection was performed, following which the eyes were stable.

One of the four patients with Nevus of Ota developed systemic metastasis and passed away 2.5 years after the diagnosis of choroidal melanoma.

Outcomes after metastasis were poor overall. Mean survival following metastasis diagnosis was 3.9 years (median: 2.5 years; range: 0.1–7.1 years). The mean overall survival after diagnosis was 11.3 years (range: 0.1–14 years). The five-year disease-specific survival (DSS) was 84% (95% CI: 77%–90%), and the 10-year DSS was 66% (95% CI: 53%–77%). In contrast, the 5-year all-cause survival was 75% (95% CI: 65%–87%), and the 10-year all-cause survival was 59% (95% CI: 47%–71%) [Fig. 1]. The overall survival probability curves and DSS curves are shown in Fig. 2.

Survival decreased with advancing AJCC T stage. The five-year survival was >90% for T1/T2, 80% for T3, and 65% for T4. The Kaplan Meier disease specific (metastasis related deaths) T stage wise overall and 5-year survival curves are shown in Figs. 3 and 4 respectively. The number at risk is provided for each time point. One patient passed away in the 11th year after developing metastasis seven years post-diagnosis.

All-cause mortality occurred in 39 patients (26%), of whom 29 (19%) died from melanoma metastasis (disease-specific mortality). The AJCC T stage-wise mortality rates and age-wise mortality rates are shown in Fig. 5. Multivariate Cox proportional hazards analysis revealed a significantly increased risk of disease-specific mortality with higher tumor stage, i.e., T2: HR = 1.78 (95% CI: 1.01–3.13; $P = 0.045$), T3: HR = 4.24 (95% CI: 2.35–7.64; $P < 0.001$) and T4: HR = 6.38 (95% CI: 3.21–12.67; $P < 0.001$), using T1 as reference.

Additional univariate predictors of increased disease-specific mortality included peripapillary tumor location obscuring or <2 mm from ONH ($P = 0.013$), basal diameter ($P = 0.049$), histopathologic scleral involvement ($P = 0.002$), abnormal chest radiographs at diagnosis ($P < 0.001$), and type of enucleation (primary vs. secondary): patients undergoing primary enucleation had better survival ($P < 0.001$).

In multivariate Cox regression, the following remained independently associated with increased disease-specific

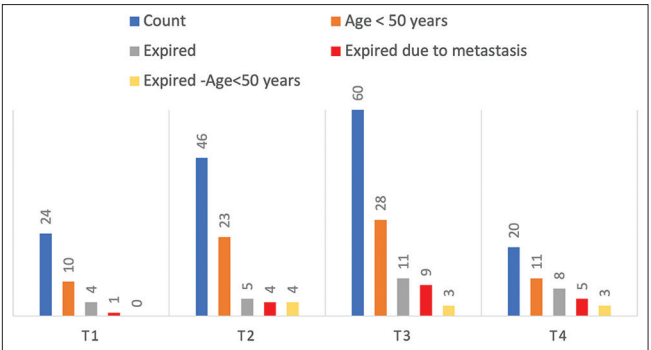


Figure 5: AJCC T-stage-wise and age-wise mortality comparison graph

mortality, i.e., greater tumor extent (HR = 3.86; 95% CI: 1.85–8.04; $P < 0.001$), secondary enucleation (HR = 1.34; 95% CI: 1.07–2.61; $P = 0.027$), and scleral involvement (HR = 3.20; 95% CI: 1.47–6.94; $P = 0.003$).

Discussion

This study contributes to the growing body of literature on uveal melanoma (UM) in the Asian Indian population, focusing on long-term survival outcomes and metastatic risk. In line with prior studies from India, including those by Biswas *et al.*, Kapoor *et al.*, and Kashyap *et al.*, we found that Indian patients tend to present at a younger age compared to Western cohorts.^[4,5,15] However, our study uniquely emphasizes survival patterns with a relatively longer mean follow-up of 66 months and includes detailed patient-level verification of survival status. Also in congruence with other Asian studies, the survival rates in the Asian population are slightly better, i.e., five-year survival rates being 75–80% [Table 5].^[12,21]

As per the Western literature, almost 50% of all patients with choroidal melanoma die of metastatic disease within 10 years, despite successful eradication of the primary ocular tumor.^[9] As mentioned by Damato *et al.*^[22] that in patients dying of metastatic disease from choroidal melanoma, the survival in our study also did not correlate significantly with any of the predictive factors for metastasis.

Our five-year metastasis-related mortality was 16% compared to 17% as mentioned by Yue *et al.* in the Chinese population, which is significantly lesser than the Caucasian studies, which is 56–61%.^[9,13,22] There could be differences in the genetic makeup of the tumor, which probably explains the better prognosis and survival in the Asian population, along with a younger onset at presentation. However, we did not find a significant difference in the risk of metastasis across different age groups. Hence, other factors like the extent of tumor, the stage of tumor, involvement of ciliary body, and sclera, as well as genetic composition of tumor, seem to have a greater association with risk of metastasis.

From the COMS natural history study, of the 45 patients who followed up in the natural history, 19 died, of which six were metastasis-related deaths.^[23] The COMS small melanoma mortality report suggested that out of the 204 patients recruited, 27 patients died, and six of them were reported to be due to metastasis-related, i.e., 2.9%.^[24] The Kaplan–Meier estimate of five-year all-cause mortality was 6.0% in small melanomas. The small fatal melanoma study showed that the median survival was six years in patients with metastatic small

Table 5: Comparison of our results with other published studies

Study	Population	Number (n)	Total deaths	Study Duration	All-cause mortality at 5 years (95% Confidence Interval)	Metastasis related deaths (% of total enrolled)
COMS Natural History Study ^[23]	White non-Hispanic	45	19	4-11Y	30% (18–47%)	8 (18%)
COMS medium melanoma Treatment Trial ^[26]	White non-Hispanic	1317 medium	364	2-13Y	18% (16–20%)	202 (15%)
COMS large melanoma Treatment Trial ^[27]	White non-Hispanic	801 analysed of 1003	457	5Y	39% (36-43%)	269 (34%)
Kujala <i>et al.</i> ^[9]	Finland (Whites)	289	239	35 Y	31% (26-37%)	145 (61%)
Stalhammer <i>et al.</i> ^[10]	Sweden (Whites)	677	533	20Y	27%	NA
Damato <i>et al.</i> ^[22]	UK	3072	1005	20Y	NA	561 (56%)
LLY Tan <i>et al.</i> ^[11]	Singapore+Chinese	25	5	1-15Y	23.4%	NA
Yue <i>et al.</i> ^[13]	Chinese	171	48	1-11Y	16% (11-24%)	30 (17.5%)
Liu <i>et al.</i> ^[14]	Chinese	582	32	1-11 Y	5.5%	29 (5%)
Kapoor AG <i>et al.</i> ^[15]	Indian	321	NA	1-24 Y	NA	16 (5%)
Meeralakshmi <i>et al.</i> ^[16]	Indian	44	4	1-7 Y	9%	3
Our study	Indian	150	39	1-15Y	19%(13-35%)	29 (16%) 10% - Medium 32% - Large

melanomas, while the Kaplan–Meier estimate of metastatic death was 52% at one year, 77% at two years, and 84% at three years after metastasis detection.^[25] In our study, there was one metastasis-related death in a patient with T1 stage who was initially treated with brachytherapy and developed a multilobular tumor recurrence one year later and underwent enucleation. The patient developed metastasis two years post-enucleation and passed away after three years of diagnosis of metastasis.

Comparing the COMS natural history cohort versus treatment trial cohort, allcause five-year mortality was 30% (42/45 were medium-sized melanoma) (95% CI, 18%–47%) in natural history cohort, while it was 18% (95% CI, 16%–20%) and 39% (36–43%) in the COMS medium melanoma treatment trial and large melanoma treatment trial, respectively.^[23,26,27] After adjusting for baseline age and longest basal diameter, the adjusted risk ratio was 1.54. Also, in COMS, patients with extrascleral extension ≥ 2.0 mm, diffuse, ring, or multifocal tumors; or tumors judged to be predominantly ciliary body melanoma were ineligible, which could be a reason for the lower mortality rate in this study compared to others.

In a Finnish study by Kujala *et al.*^[9] where all the patients were treated with radical surgery either by enucleation or exenteration, uveal melanoma-related mortality was noted to be as high as 31% and 45%, within 5 and 15 years, respectively, which is much higher compared to most other studies. In a Swedish study by Stalhammer, uveal melanoma-related mortality after brachytherapy was 18% by five years, 28% by 10 years, 32% by 15 years, 35% by 20 years, and 36% by 25 to 40 years.^[28] Our survival rates (75%) are similar to those of another multi-ethnic Asian study, which reported 76.6%.^[11] A comparative study between Chinese and American cohorts concluded that compared with American patients, Chinese patients had younger onset age and larger tumors at diagnosis

and better prognosis, mainly because of their younger age, which is also seen in our study.^[12]

Local recurrence is associated with an increased risk of dying of melanoma. In a study by Lane *et al.*,^[29] the median time from initial treatment to recurrence was 30.5 months, and the five-year probabilities of melanoma-related mortality were 9.5% in patients without local recurrences compared with 32.0% in patients with recurrences. In our study, four of the 13 patients who had recurrence passed away.

A study by Jensen *et al.*, on Danish patients showed that the peak incidence of metastasis after enucleation occurred within the first year, and over half of those who developed metastasis were within the first three years.^[30] Parallel to the results published in previous studies, the annual rate of metastasis decreases over time since diagnosis and peaks at approximately three to four years after diagnosis. Our study also showed that most metastases (24/29–83%) occur within the first five years.^[31]

A multicentric international data-registry-based study showed that of the 69 patients who presented with metastasis at initial presentation (stage IV–AJCC), five patients (9%) were T1, and nearly 30% each belonged to T2, T3, and T4, respectively.^[32] In our study, we had two patients with metastasis at presentation, one with T3a who had bone metastasis at presentation, and passed away within five months of diagnosis, and another with T3d with brain metastasis at presentation, who passed away 3.3 years after diagnosis.

Strengths of this study include a relatively long follow-up period, comprehensive clinical documentation of all parameters, assessing all perspectives from globe salvage to life salvage, including both COMS and AJCC TNM classification, and verification of patient outcomes through direct contact. This is important in a setting like India, where we do not have national death registries/cancer-related death registries, unlike the Western

world, to know the survival status. However, it is important to note that other Indian studies have reported outcomes in UM using standard follow-up protocols, although not through direct contact.

Limitations of our study include its retrospective single-center design, lack of genetic or molecular tumor profiling (e.g., GEP and PRAME), and the absence of standardized systemic therapy protocols for metastasis management. Additionally, while we attempted exhaustive follow-up, some degree of survival misclassification may remain due to patient recall or incomplete information from family members. Also availability of genetic testing and patient cost factors limits the tumor genetic/molecular profiling in our population.

So far, only two studies have been published reporting survival rates from India: One by Meeralakshmi *et al.*,^[16] on survival outcomes comparing brachytherapy and enucleation, which did not reveal any difference between the two treatment modalities, although the sample size was very small. The metastasis-free survival was 95% in the brachytherapy group and 91% in the enucleation group in this study. Another study by Kapoor *et al.*^[15] reported the five-year survival to be 97.5% (T1, T2 – 100% while in T3 – 98.3% and T4 – 92.3%). Both these studies did not report on tracking the patients or checking any death registries to study the survival status specifically, and the mean follow-up duration was short, i.e., 39 and 27 months, respectively. In our study, we have confirmed the survival status of every patient by contacting them or their kin. This may be the reason that we found slightly higher mortality rates in our study and a significant difference between outcomes and metastasis rates across different T stages, with a longer mean follow-up of 66 months.

Conclusion

This study sheds light on the clinical outcomes of choroidal melanoma in the Indian population, revealing that, despite a younger age at presentation, survival outcomes are favorable. Early detection and monitoring during the first five years post-treatment are vital, and structured follow-ups should be part of the clinical management strategy. The authors would like to propose a national database/registry for UMs in Indian patients, along with a preferred practice consensus with genetic testing, which would help in a better understanding of the factors influencing survival and metastasis in Indian patients with choroidal melanoma, and plan structured surveillance and treatment strategies.

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